

Project Report

Title: Biclustering Gene Expression Data Using Hierarchical BiClustering
Bioinformatics (2025)

1. Introduction

The goal of this project is to investigate how biclustering and deep-learning-based denoising can be combined to uncover structure in a real gene expression dataset. The dataset is the Spellman yeast cell-cycle expression matrix, containing thousands of genes profiled at multiple time points across the cell cycle. Each row corresponds to a gene, and each column (except the first) corresponds to a time point.

Biclustering simultaneously groups genes and time points to identify submatrices (biclusters) where both rows and columns exhibit coherent expression patterns. In contrast to conventional clustering, which clusters only genes or only samples, biclustering can reveal local patterns confined to subsets of time points.

This project implements two models:

- **Model 1 (Raw data):** Hierarchical biclustering and spectral biclustering directly on a normalized version of the Spellman matrix.
- **Model 2 (AE-denoised data):** A denoising autoencoder (AE) is trained on the same normalized matrix. Biclustering is then repeated on the AE-reconstructed (denoised) matrix.

Both models are evaluated using Mean Squared Residue (MSR) and variance of each bicluster, as well as visual inspection of heatmaps and reordered matrices. The main questions are:

1. Does AE-based denoising increase or reduce the number of valid biclusters?
2. Does the MSR improve after denoising?
3. How do the reordered heatmaps differ between raw and denoised data?

2. Data Preprocessing

2.1 Dataset loading and structure

The Spellman dataset (Spellman.csv) was loaded in Cell 2. The first column, labeled time, contains gene identifiers (e.g., YAL001C), and the remaining 23 columns correspond to time points (40, 50, ..., 260 minutes).

- Raw data shape (rows, columns): (4381, 24)
- Number of genes: 4381
- Number of time points: 23
- Time points (ordered): ['40', '50', ..., '260']

The numeric expression matrix used for analysis is:

- Expression matrix shape (genes $\times$ timepoints): (4381, 23)

A basic histogram of all raw expression values is shown in:

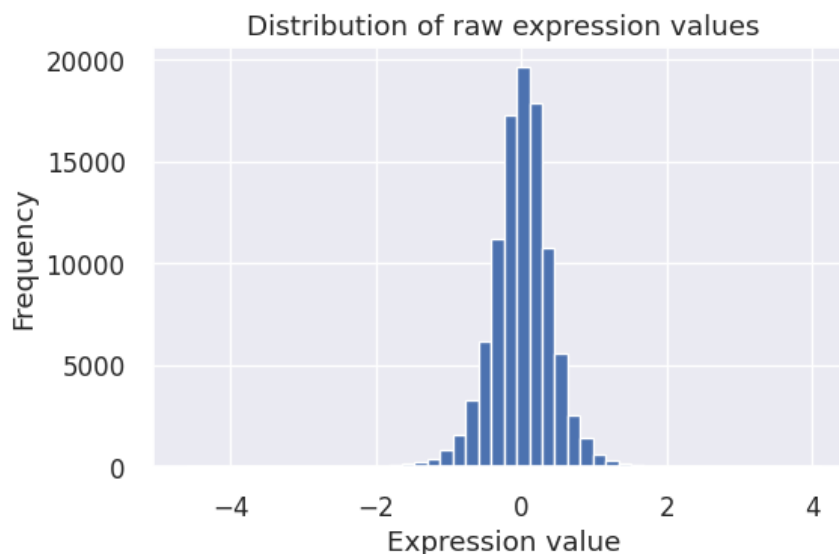


Figure 1 – “Distribution of raw expression values”

This histogram is approximately bell-shaped and centered around zero, with most values between about -2 and +2, but with tails extending to about -4.6 and +4.1.

Global statistics on the raw values (Cell 3):

- Min: -4.63
- Max: +4.14
- Mean: 0.0017
- Std. dev.: 0.4245

These statistics confirm a roughly centered but moderately heavy-tailed distribution.

2.2 Missing values and data quality

In Cell 3, the matrix was checked for numerical issues:

- Number of NaN values: 0 (0.0% of entries)
- Number of Inf values: 0 (0.0% of entries)

Therefore, no imputation or removal of entries based on NaN/Inf was necessary, and the original numeric matrix could be used directly for further preprocessing.

2.3 Normalization

To make genes comparable and to remove global scale differences, each row (gene) was standardised to have mean 0 and variance 1 across time points (row-wise z-score normalization). The normalised matrix maintains the same shape as the original numeric matrix:

- Normalised matrix shape: (4381, 23)

Checks on the first 5 genes confirmed:

- Row-wise means are approximately 0 (on the order of 10^{-8}).
- Row-wise standard deviations are ≈ 1 .

This normalisation makes the expression profiles comparable across genes and stabilises the training of the autoencoder as well as the behaviour of distance-based clustering methods.

2.4 Variance-based gene filtering

Genes with very flat expression across time contain little information for clustering and tend to add noise. To remove such genes, the row-wise variance distribution was computed on the raw data:

- Min variance: 0.0116
- 10th percentile: 0.0527
- Median: 0.1243
- 90th percentile: 0.3334

Based on this distribution, the lowest 20% of genes by variance were removed (Cell 4):

- Filtering strategy: remove lowest 20% of genes by row variance
- Variance threshold (20th percentile): 0.0697
- Removed genes: 877 / 4381
- Remaining genes: 3504

The filtered, normalised matrix used in all subsequent analyses is thus:

- Final analysis matrix shape: (3504 genes, 23 time points)

A preview of this matrix is shown in:

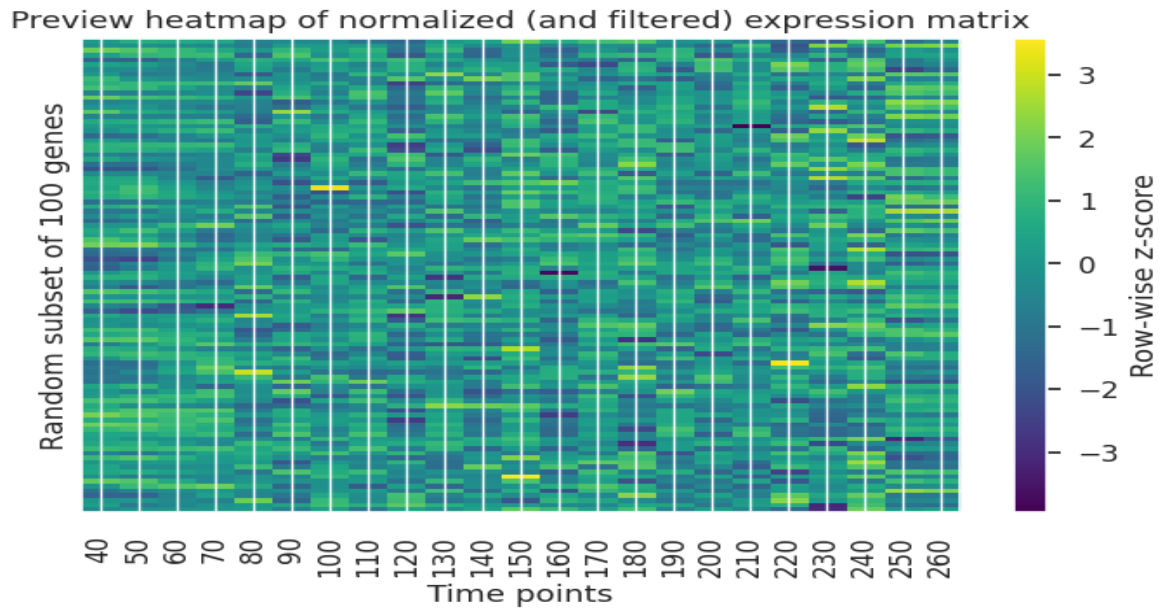


Figure 2 – “Preview heatmap of normalized (and filtered) expression matrix”

This figure suggests heterogeneous temporal patterns and confirms that the variance-based filtering retains substantial biological variability while discarding uninformative, nearly constant genes.

3. Model Implementation

3.1 Biclustering pipeline

All biclustering analyses in Part 1 were applied to the normalised and variance-filtered expression matrix of size 3504×23 (genes $\times$ time points). Two biclustering algorithms were used:

- Hierarchical biclustering (AgglomerativeClustering with Ward linkage)
- Spectral biclustering (sklearn SpectralBiclustering)

In both cases, biclustering is implemented as separate clustering of rows and columns:

- Cluster the genes into k_r row clusters.
- Cluster the time points into k_c column clusters.
- Each (row_cluster r , col_cluster c) pair defines one bicluster, i.e. a rectangular submatrix of the expression matrix.

To choose the number of clusters, a silhouette-based model selection step (Cell 7) was performed on the normalised matrix X using candidate values $k \in \{3, \dots, 9\}$ for both rows and columns. The silhouette scores suggested:

- Best number of gene clusters (rows): $k_r = 3$
- Best number of time-point clusters (columns): $k_c = 9$

These values were then fixed for all subsequent analyses in Part 1, providing a good compromise between cluster resolution and interpretability of the resulting heatmaps. With $(k_r, k_c) = (3, 9)$, each method produces:

- Total number of biclusters per method: $k_r \times k_c = 27$

Each bicluster is evaluated using two quantitative measures:

- MSR (Mean Squared Residue):
 - Measures how well the bicluster can be explained by additive row and column effects.
 - Lower MSR indicates a more coherent 2D pattern (genes and time points vary together in a consistent way).
- Variance:
 - Measures how dynamic the expression values are within the bicluster.
 - High variance indicates a strong signal; very low variance may correspond to a trivially “flat” but coherent pattern.

To classify biclusters as “valid” versus “invalid”, a single global MSR threshold was defined from the combined MSR distribution of all biclusters obtained on the raw normalised data (both hierarchical and spectral). Specifically:

- MSR threshold = 35th percentile of the combined MSR values

- Biclusters with $MSR \leq \text{threshold}$ are labelled as valid; the rest are labelled as invalid

This unified threshold allows a fair comparison between methods (hierarchical vs spectral) and later between models (raw vs AE-denoised data).

3.1.1 Hierarchical biclustering

For hierarchical biclustering, genes and time points were clustered separately using AgglomerativeClustering with Ward linkage and Euclidean distance on the normalised matrix X:

- Row clustering (genes):
 - Input: each gene is represented by its 23-dimensional z-scored time-course.
 - Algorithm: AgglomerativeClustering, linkage = "ward".
 - Output: assignment of all genes into $k_r = 3$ clusters.
- Column clustering (time points):
 - Input: the transposed matrix X^T , so that each time point is represented by a 3504-dimensional vector of gene expressions.
 - Algorithm: AgglomerativeClustering, linkage = "ward".
 - Output: assignment of all 23 time points into $k_c = 9$ clusters.

The matrix is then reordered by sorting rows and columns according to their cluster labels, resulting in a block-structured heatmap in which coherent biclusters appear as approximately rectangular blocks (Cell 9). These row and column labels are subsequently used to:

- enumerate the 27 biclusters as all (row_cluster, col_cluster) pairs
- compute MSR and variance for each bicluster
- determine which biclusters are valid under the global MSR threshold:

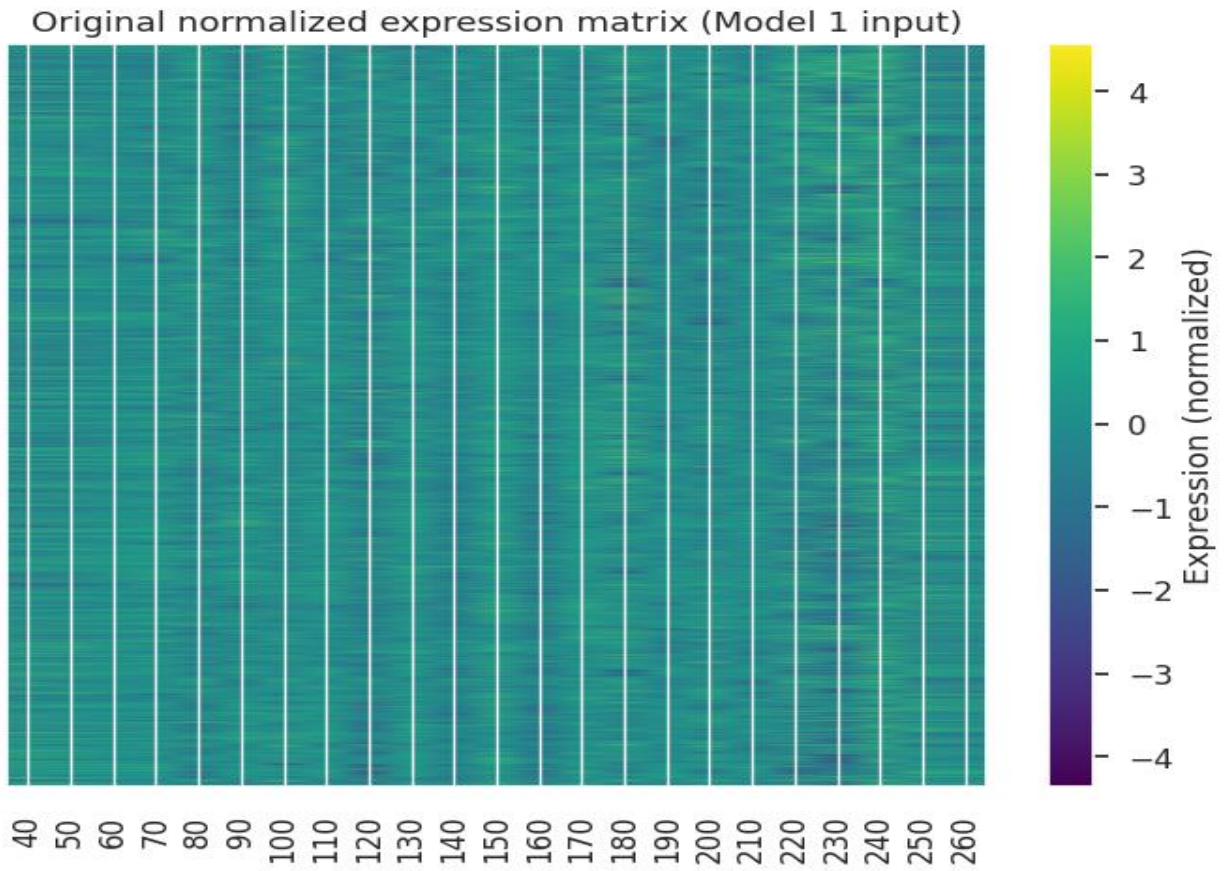


Figure 4 – “Original normalized expression matrix (Model 1 input)”

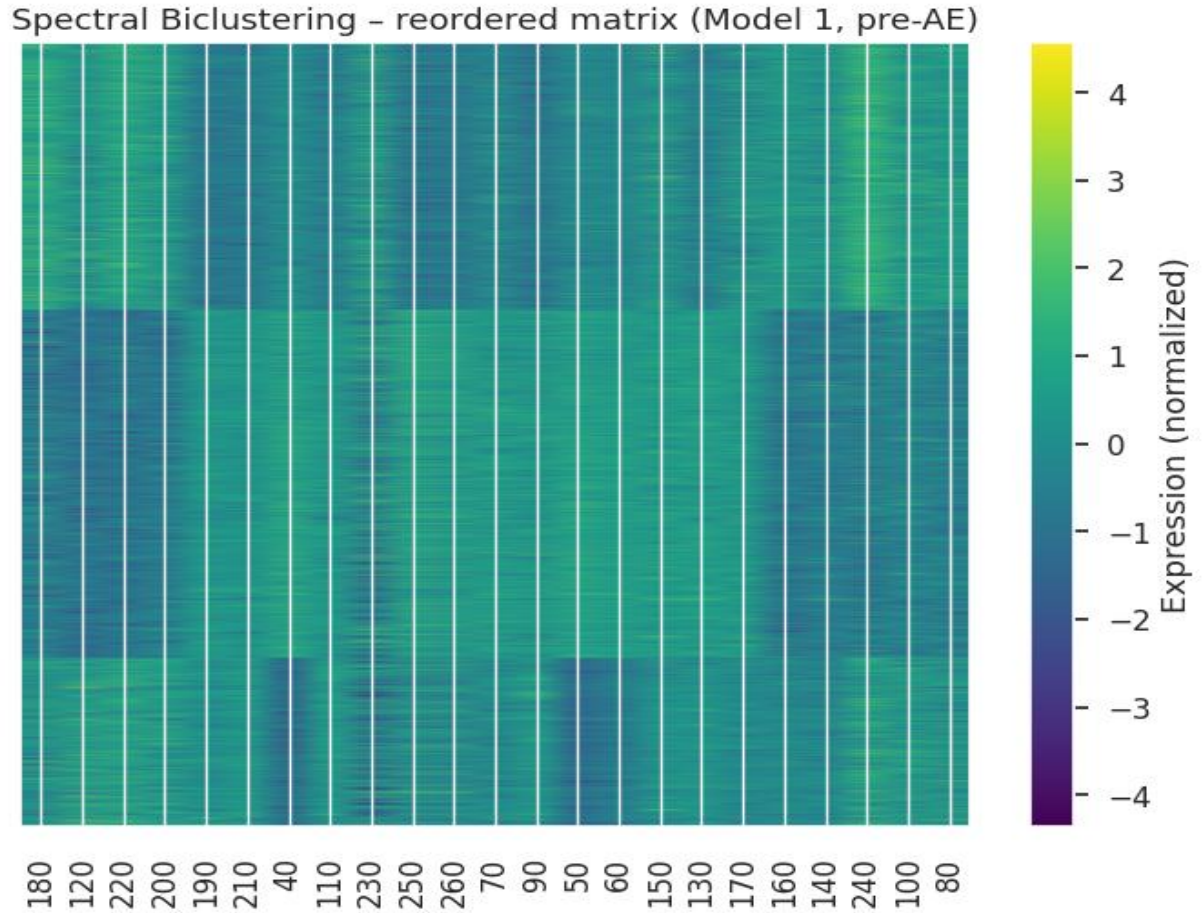


Figure 5 – “Hierarchical biclustering – reordered matrix (Model 1, pre-AE)”

The latter highlights block-like structures corresponding to biclusters.

3.1.2 Spectral biclustering

Spectral biclustering uses singular vectors of the data matrix to simultaneously cluster rows and columns. Using the same numbers of clusters:

- Row clusters: k_r
- Column clusters: $k_c = 9$

The reordered heatmap is shown in:

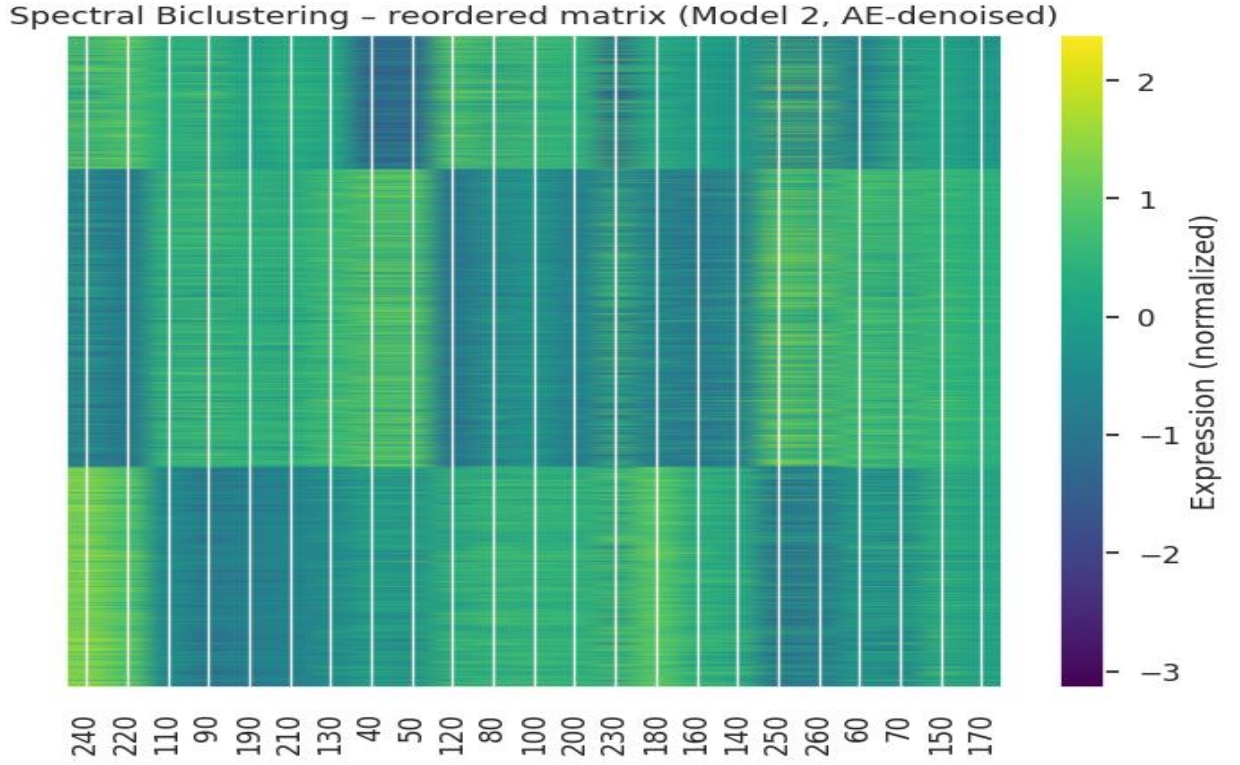


Figure 6 – “Spectral Biclustering – reordered matrix (Model 1, pre-AE)”

This representation often reveals checkerboard-like structures reflecting coherent biclusters.

3.1.3 Bicluster extraction and scoring

From both methods, all $k_r \times k_c = 3 \times 9 = 27$ biclusters were extracted. For each bicluster:

- MSR and variance were computed.
- Biclusters with MSR below the 35th percentile threshold were labeled valid.

The top 5 hierarchical and spectral biclusters (sorted by lowest MSR) are visualized as heatmaps:

Hierarchical (raw) BC (row=0, col=6) | rows=875, cols=2
MSR=0.0257, Var=0.6905

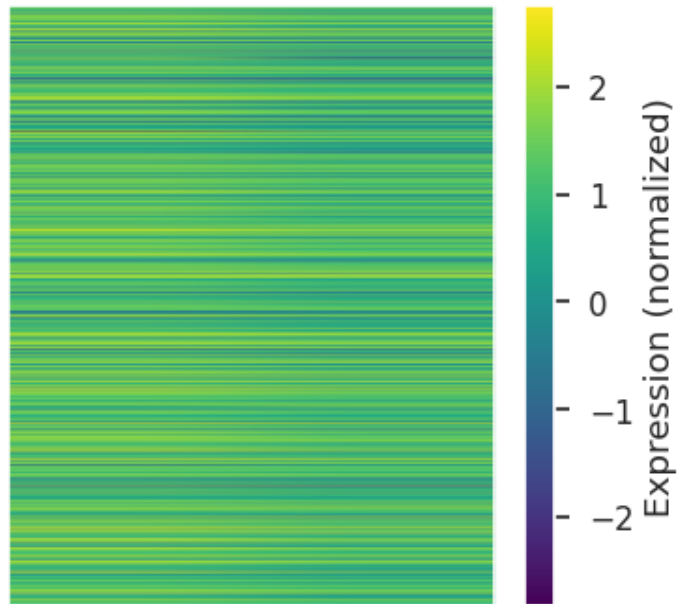


Figure 7 – Selected hierarchical biclusters (raw data)

Hierarchical (raw) BC (row=2, col=6) | rows=1091, cols=2
MSR=0.0335, Var=0.6219

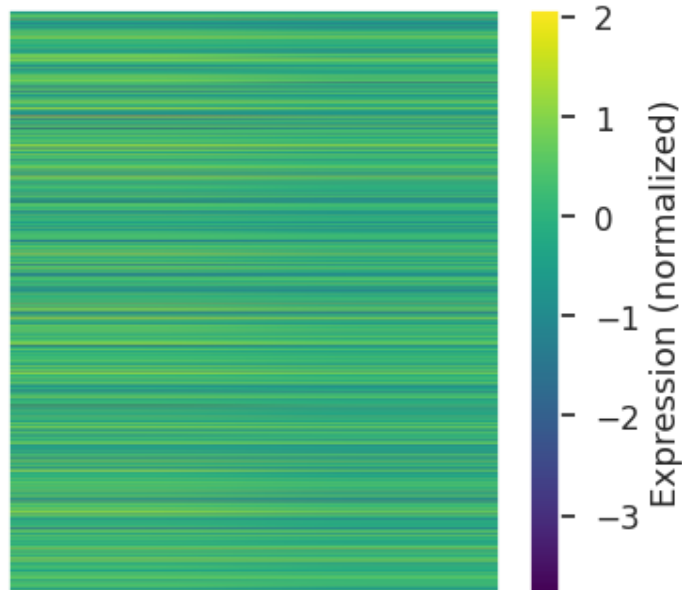


Figure 8 – Selected hierarchical biclusters (raw data)

Hierarchical (raw) BC (row=1, col=6) | rows=1538, cols=2
MSR=0.0402, Var=0.8226

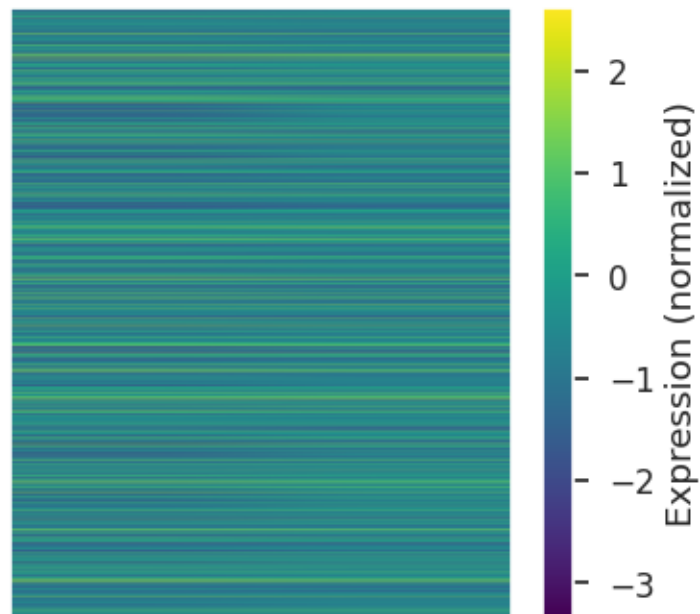


Figure 9 – Selected hierarchical biclusters (raw data)

Hierarchical (raw) BC (row=0, col=8) | rows=875, cols=2
MSR=0.1569, Var=0.8454

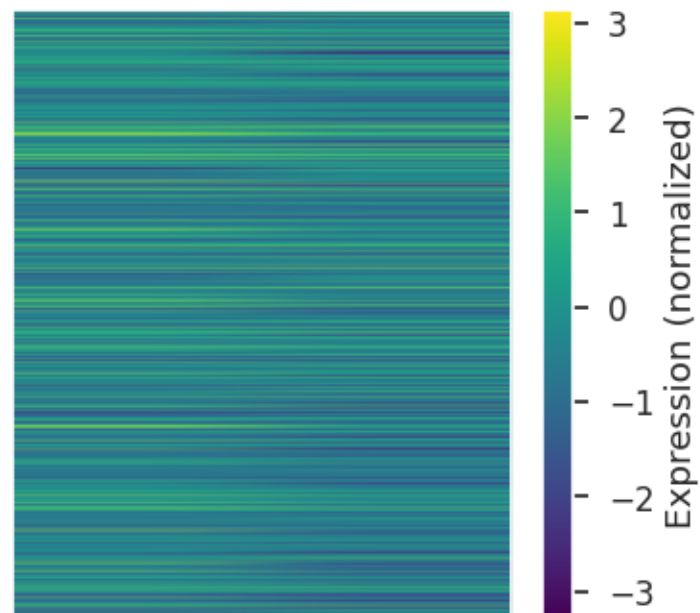


Figure 10 – Selected hierarchical biclusters (raw data)

Hierarchical (raw) BC (row=0, col=0) | rows=875, cols=4
MSR=0.1997, Var=0.6760

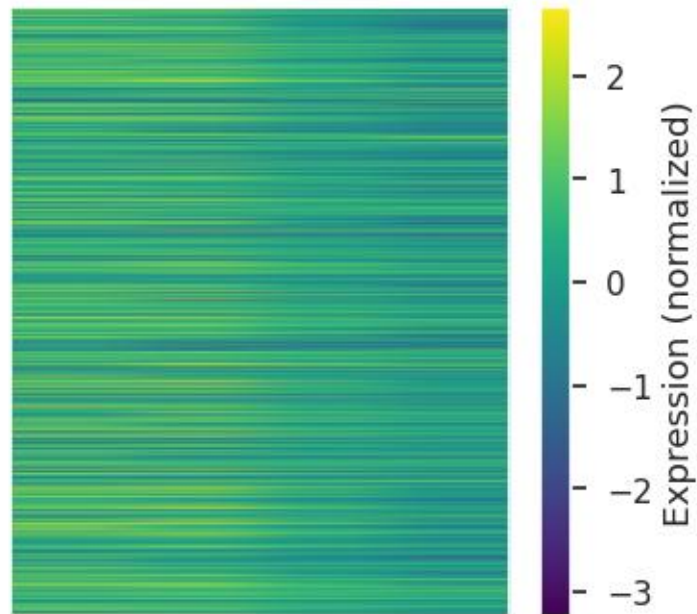


Figure 11 – Selected hierarchical biclusters (raw data)

Spectral (raw) BC (row=1, col=3) | rows=1551, cols=2
MSR=0.0243, Var=0.7615

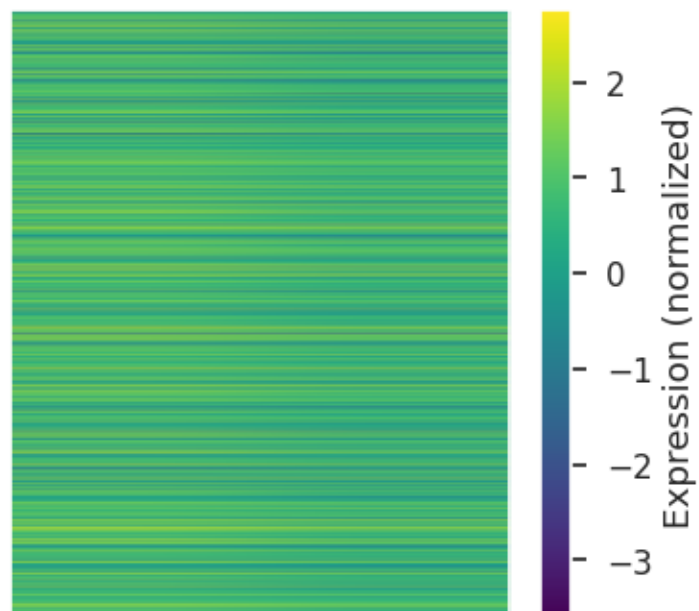


Figure 12 – Selected spectral biclusters (raw data)

Spectral (raw) BC (row=2, col=3) | rows=753, cols=2
MSR=0.0249, Var=1.0431

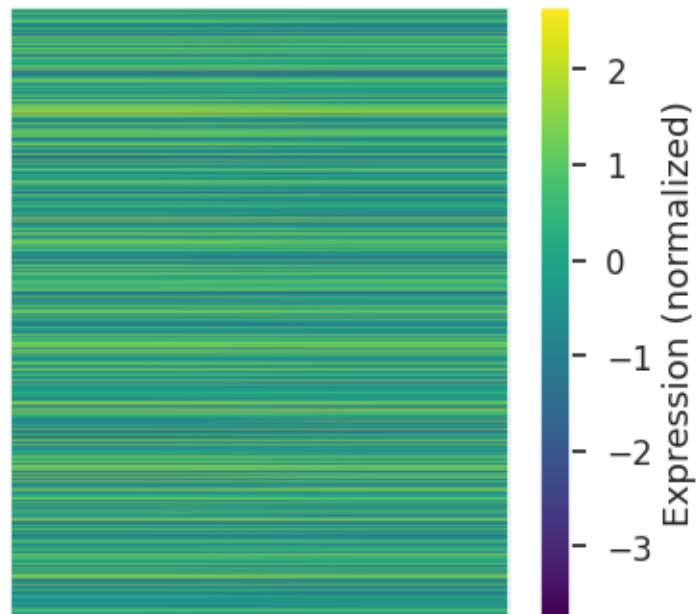


Figure 13 – Selected spectral biclusters (raw data)

Spectral (raw) BC (row=0, col=3) | rows=1200, cols=2
MSR=0.0512, Var=0.6262

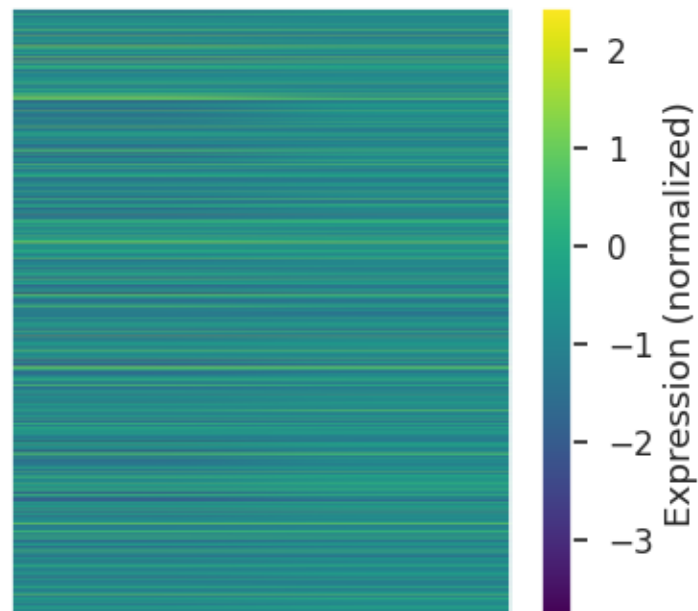


Figure 14 – Selected spectral biclusters (raw data)

Spectral (raw) BC (row=2, col=6) | rows=753, cols=2
MSR=0.1937, Var=0.6179

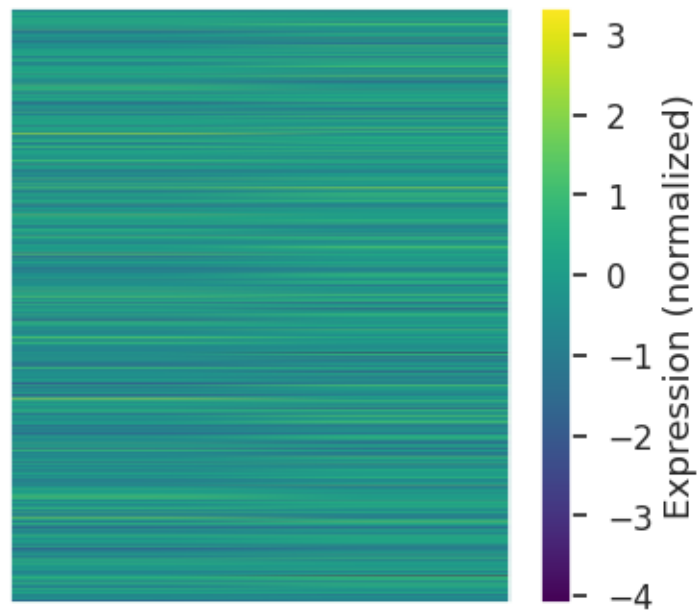


Figure 15 – Selected spectral biclusters (raw data)

Spectral (raw) BC (row=1, col=6) | rows=1551, cols=2
MSR=0.2028, Var=0.6821

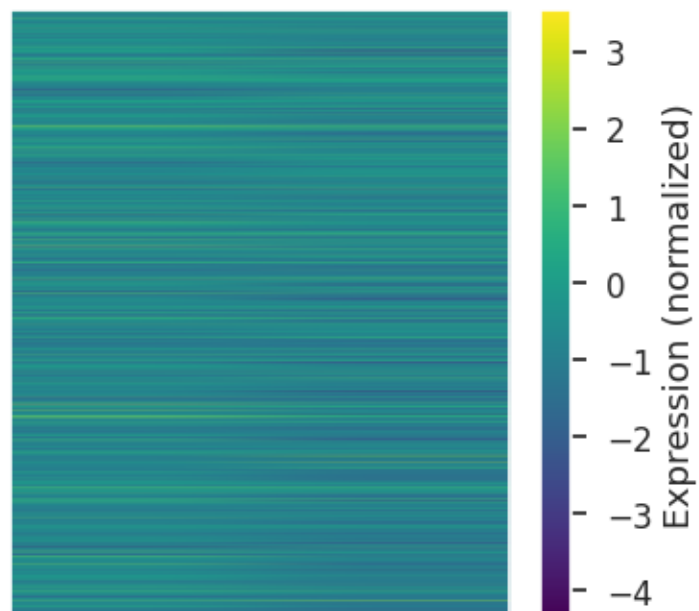


Figure 16 – Selected spectral biclusters (raw data)

These figures help visually assess whether biclusters capture meaningful temporal patterns.

3.2 Denoising autoencoder (Model 2)

To reduce noise and enhance underlying structure, a denoising autoencoder was trained on the same normalized, filtered matrix (3723×23).

3.2.1 Architecture

The denoising autoencoder (AE) in Version 2 (Cells 13–14) operates on the normalised, variance-filtered expression matrix X of size 3504×23 (genes $\times$ time points). Each gene is represented by a 23-dimensional z-scored time-course. The AE architecture is:

- Input layer: 23-dimensional gene expression vector (one time-course per gene)
- Gaussian noise layer: std = 0.20 (applied only during training)
- Encoder hidden layer: Dense, 32 units, ReLU activation
- Latent (bottleneck) layer: Dense, 3 units, ReLU activation
- Decoder hidden layer: Dense, 32 units, ReLU activation
- Output layer: Dense, 23 units, linear activation (reconstruction)

The Keras model summary reports:

- Input dimension: 23
- Latent dimension: 3 (≈ 7 – $8\times$ compression of the 23 time points)
- Hidden width: 32 units in both encoder and decoder
- Total parameters: 1,754

This is a compact network, deliberately kept small to promote meaningful compression and denoising while limiting over-parameterisation and training time.

3.2.2 Rationale for architectural choices

Latent dimension = 3

Using a 3-dimensional bottleneck imposes a relatively strong compression on the 23-dimensional input (roughly 7 – $8\times$). The intention is that the AE captures only the

dominant temporal modes of variation (e.g. broad cell-cycle phases or global activation/repression patterns) and discards high-frequency noise and gene-specific idiosyncrasies.

Empirically, the trained model achieves a global reconstruction error of:

- Mean per-gene reconstruction MSE ≈ 0.40

with per-gene MSE values mostly between ~ 0.24 and ~ 0.54 . This indicates that the reconstruction is reasonably faithful while still enforcing noticeable smoothing, which is desirable before biclustering.

Hidden dimension = 32

Each side of the bottleneck uses a single hidden layer with 32 ReLU units. This provides sufficient non-linear capacity to model moderately complex temporal trajectories (e.g. non-monotonic or multi-peak patterns) without making the model unnecessarily deep or wide. A larger hidden dimension would increase capacity and risk overfitting, while a much smaller one might underfit and oversimplify the profiles.

Gaussian noise std = 0.20

A GaussianNoise layer with standard deviation 0.20 is applied to the inputs during training. Because the data are row-wise z-scored (per-gene mean 0, variance 1), this noise level represents a moderate perturbation: large enough to disrupt small fluctuations but not so large as to destroy the underlying pattern. This forces the AE to learn robust features that can reconstruct the clean signal from noisy inputs, thereby acting as a genuine denoising model rather than simply memorising the input.

Training configuration

The AE is trained with:

- Optimiser: Adam, learning rate 1×10^{-3}
- Loss: mean squared error (MSE)
- Batch size: 64

- Maximum epochs: 300, with early stopping (patience = 15) and ReduceLROnPlateau

Early stopping halted training around epoch 191, restoring the weights from the best validation epoch (epoch 176), where the validation MSE reached ≈ 0.397 . This indicates stable training with no severe overfitting.

3.2.3 Biclustering on denoised data

After training, the AE was applied to all 3504 genes, producing a denoised expression matrix X_{denoised} with the same shape as the input:

- Denoised matrix shape: (3504 genes, 23 time points)

The same biclustering pipeline from Part 1 was then re-applied to X_{denoised} :

- Hierarchical biclustering (AgglomerativeClustering, Ward linkage) on rows and columns separately
- Spectral Biclustering (sklearn SpectralBiclustering with method="bistochastic")

Crucially, the clustering hyperparameters were kept identical to those used on the raw normalised data to enable a fair comparison:

- Number of gene clusters: $k_r = 3$
- Number of time-point clusters: $k_c = 9$
- Total biclusters per method: 27

For both methods, the same evaluation procedure was used as in Part 1:

- MSR (Mean Squared Residue) and variance computed for each of the 27 biclusters
- The identical global MSR threshold derived from the raw-data biclusters (35th percentile of the combined MSR distribution) was reused to classify biclusters on X_{denoised} as “valid” (MSR below threshold) or “invalid”

This design ensures that any differences in the number of valid biclusters, mean MSR, and variance between Model 1 (raw) and Model 2 (AE-denoised) can be attributed to the denoising step rather than to changes in clustering or evaluation settings.

The key figures are:

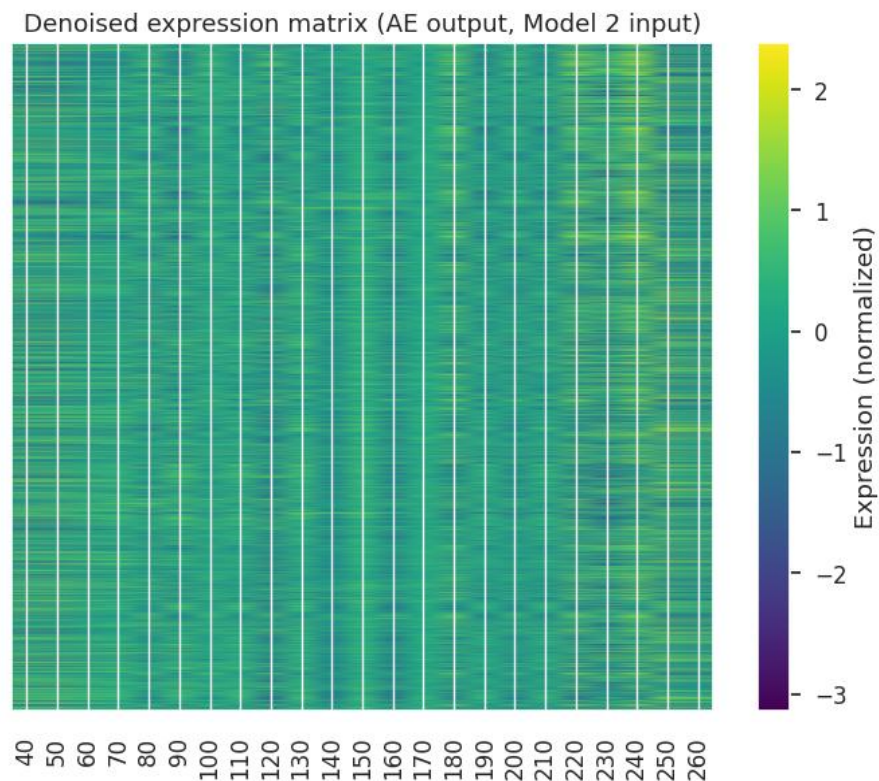


Figure 19 – “Denoised expression matrix (AE output, Model 2 input)”

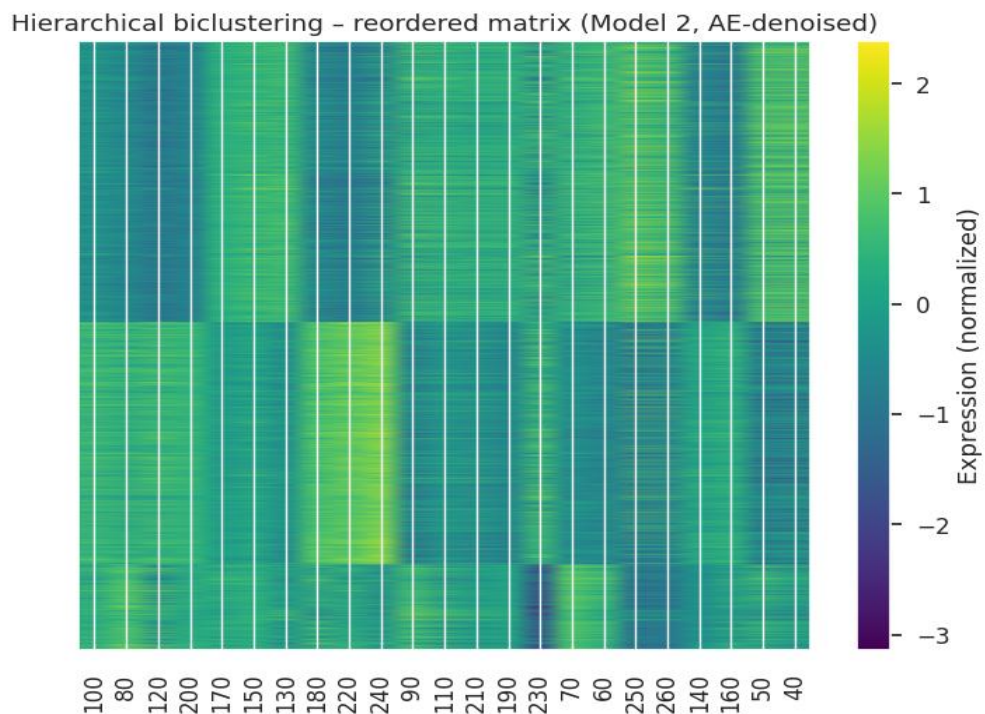


Figure 21 – “Hierarchical biclustering – reordered matrix (Model 2, AE-denoised)”

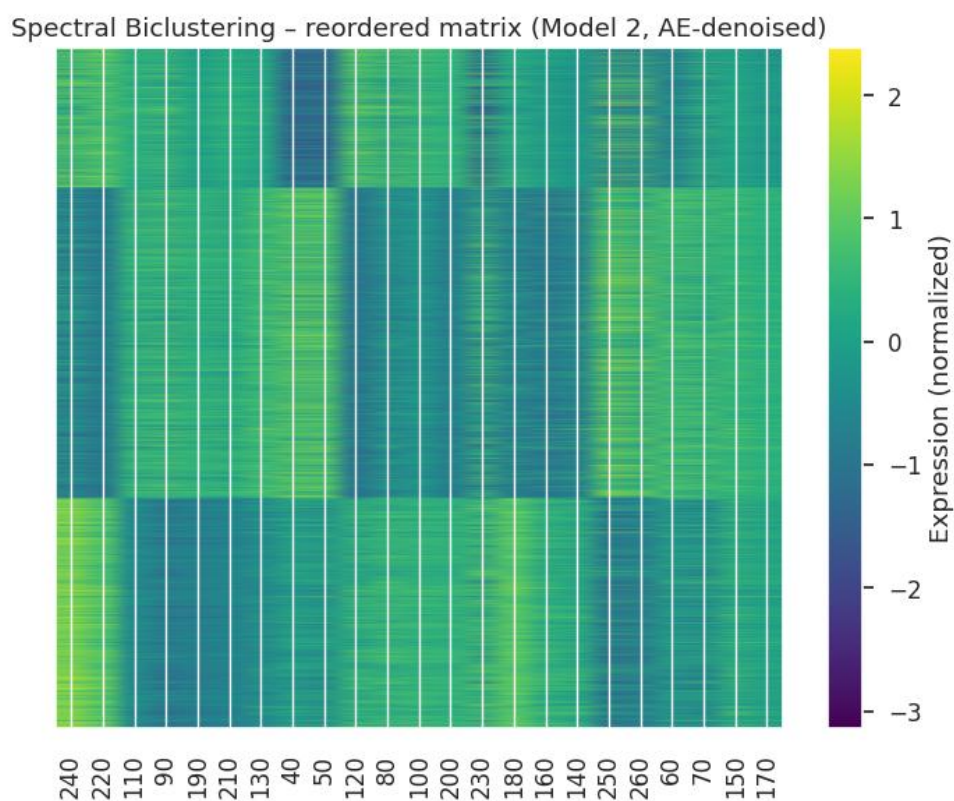


Figure 22 – “Spectral Biclustering – reordered matrix (Model 2, AE-denoised)”

Hierarchical (denoised) BC (row=0, col=6) | rows=1620, cols=2
MSR=0.0018, Var=0.4720

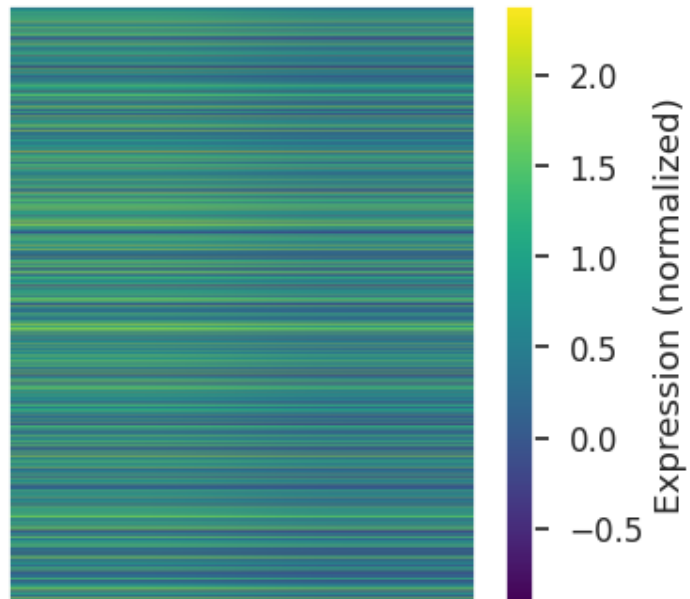


Figure 23 – Selected hierarchical biclusters (denoised)

Hierarchical (denoised) BC (row=2, col=6) | rows=492, cols=2
MSR=0.0043, Var=0.2392

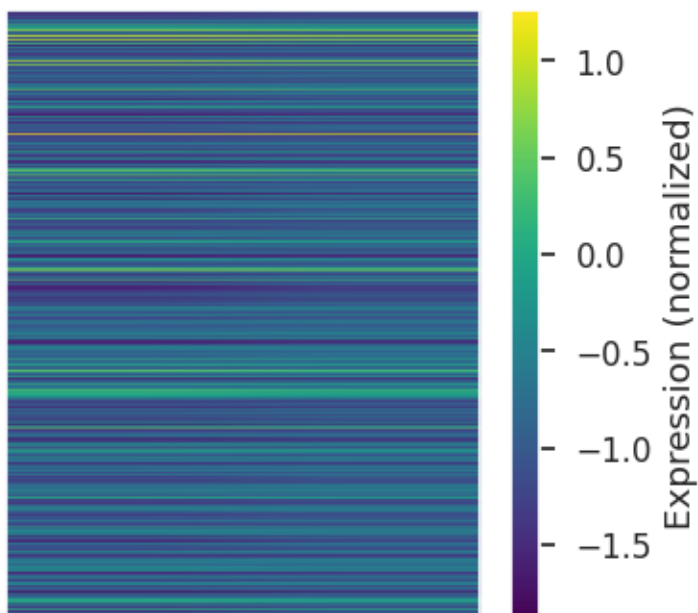


Figure 24 – Selected hierarchical biclusters (denoised)

Hierarchical (denoised) BC (row=1, col=8) | rows=1392, cols=2
MSR=0.0043, Var=0.5017

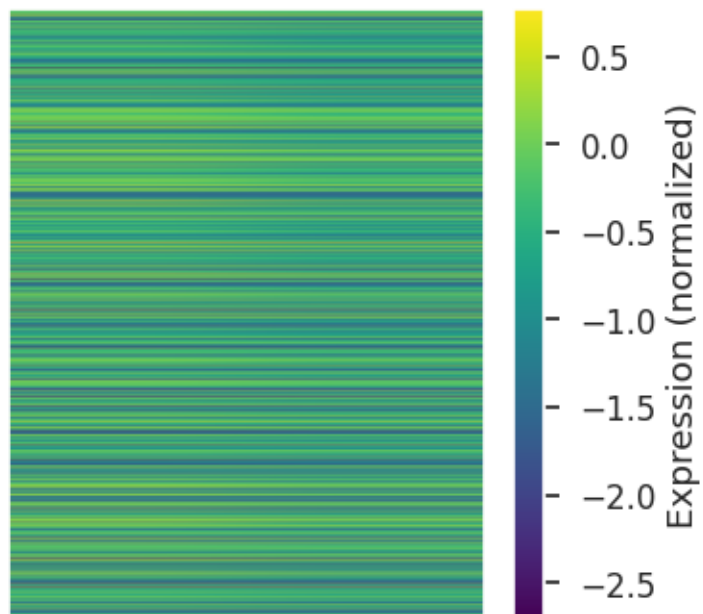


Figure 25 – Selected hierarchical biclusters (denoised)

Hierarchical (denoised) BC (row=1, col=8) | rows=1392, cols=2
MSR=0.0043, Var=0.5017

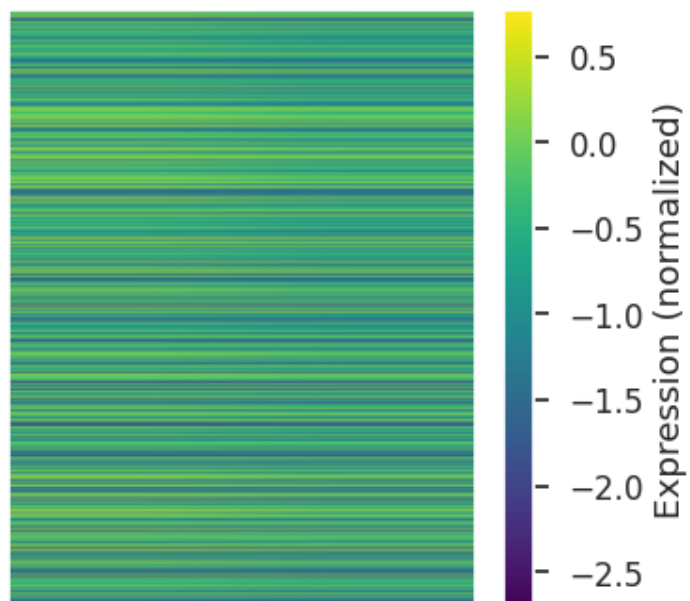


Figure 26 – Selected hierarchical biclusters (denoised)

Hierarchical (denoised) BC (row=1, col=6) | rows=1392, cols=2
MSR=0.0056, Var=0.6348

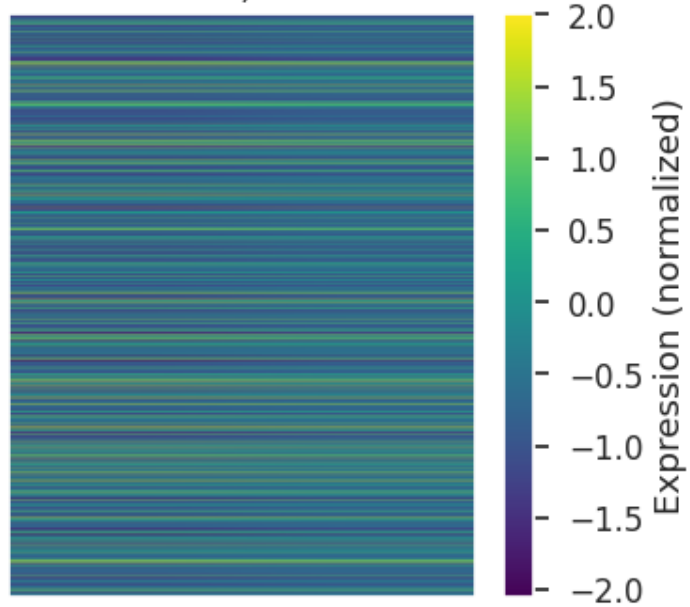


Figure 27 – Selected hierarchical biclusters (denoised)

Spectral (denoised) BC (row=0, col=6) | rows=726, cols=2
MSR=0.0019, Var=0.9192

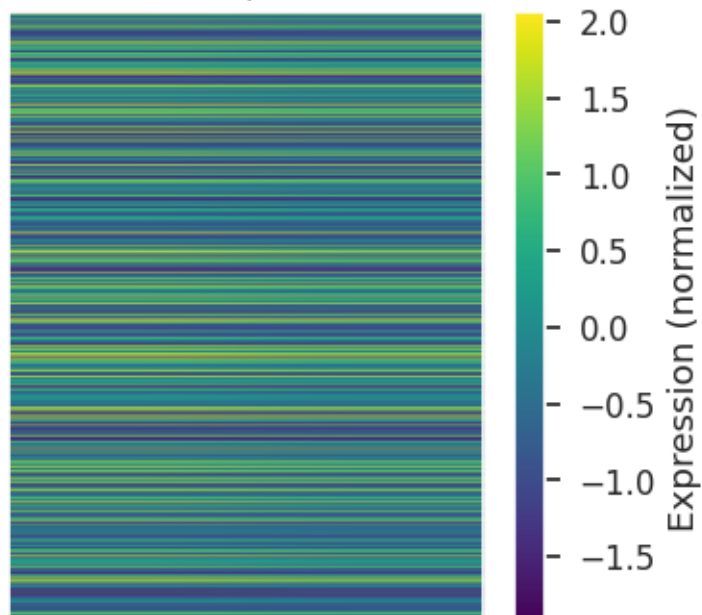


Figure 28 – Selected spectral biclusters (denoised)

Spectral (denoised) BC (row=1, col=6) | rows=1594, cols=2
MSR=0.0025, Var=0.5595

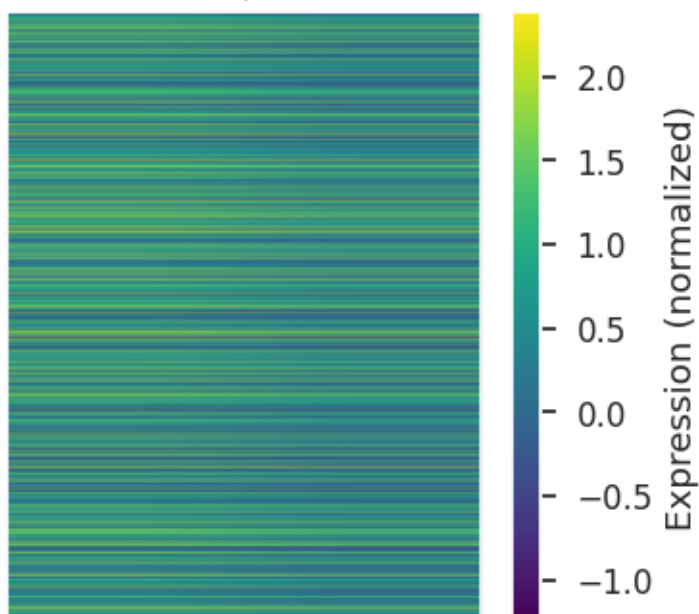


Figure 29 – Selected spectral biclusters (denoised)

Spectral (denoised) BC (row=0, col=2) | rows=726, cols=2
MSR=0.0030, Var=0.4397

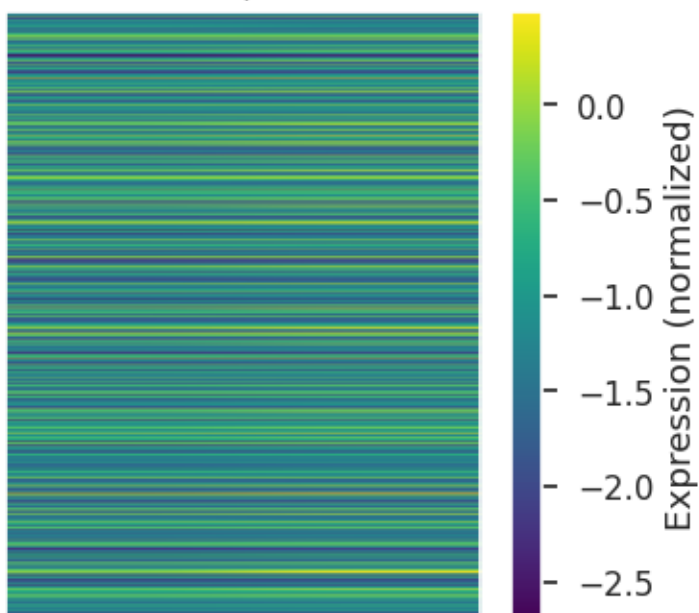


Figure 30 – Selected spectral biclusters (denoised)

Spectral (denoised) BC (row=2, col=6) | rows=1184, cols=2
MSR=0.0064, Var=0.3948

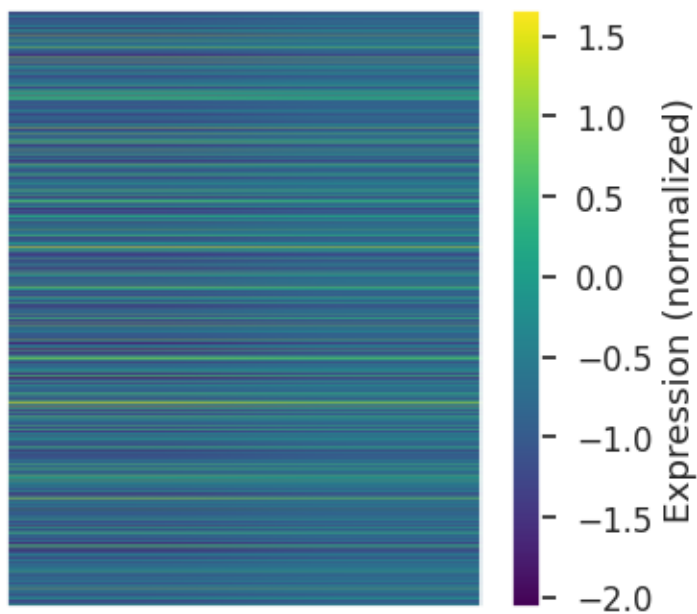


Figure 31 – Selected spectral biclusters (denoised)

Spectral (denoised) BC (row=2, col=6) | rows=1184, cols=2
MSR=0.0064, Var=0.3948

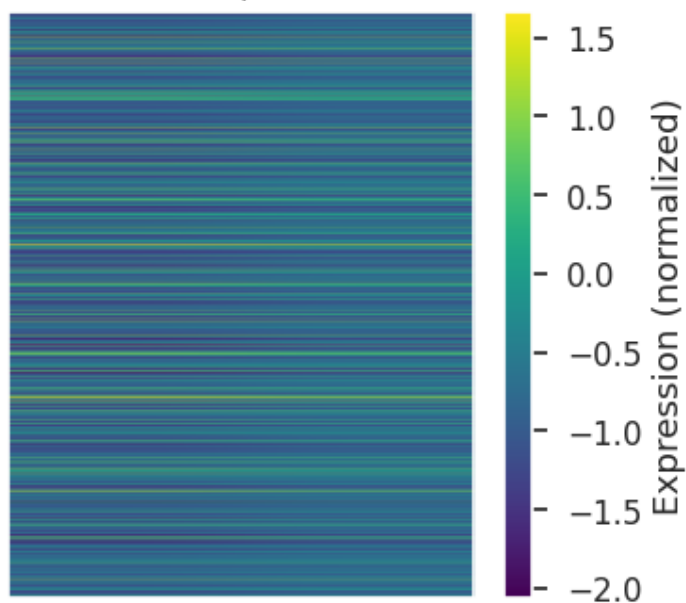


Figure 32 – Selected spectral biclusters (denoised)

Spectral (denoised) BC (row=1, col=2) | rows=1594, cols=2
MSR=0.0071, Var=0.4194

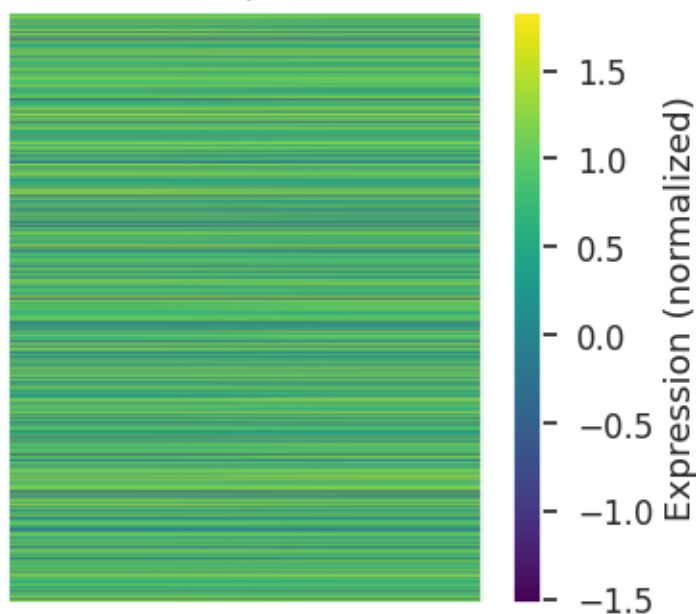


Figure 33 – Selected spectral biclusters (denoised)

4. Training Configuration and Hyperparameter Tuning

4.1 Global configuration

For reproducibility, random seeds were fixed:

- SEED = 42 for both NumPy and TensorFlow.

Library versions (Cell 1):

- NumPy: 2.0.2
- Pandas: 2.2.2
- Scikit-learn: 1.6.1
- TensorFlow: 2.19.0

GPU detected (1 device(s)) and will be used by TensorFlow.

4.2 Choosing the number of row,column clusters k_r, k_c

The number of row clusters was tuned using silhouette analysis on a random subset of 5000 genes (Cell 7). For each candidate $k_r \in \{3,4,5,6,7,8,9\}, k_c \in \{3,4,5,6,7,8,9\}$, hierarchical clustering was applied and the silhouette score was computed:

Table 1 – Silhouette scores for row clustering

k_r	Silhouette score
3	0.120
4	0.104
5	0.074
6	0.068
7	0.059
8	0.042
9	0.041

- Best score: 0.12 at $k_r = 3$.

The corresponding plot is:

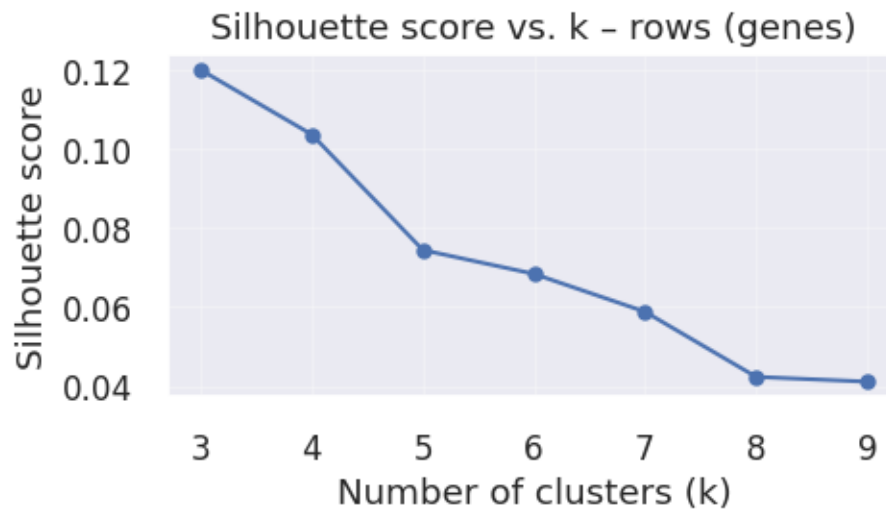


Figure 3 – “Silhouette score vs. k_r (rows)”

Table 5 – Silhouette scores for row clustering

k_c	Silhouette score
3	0.163
4	0.171
5	0.171
6	0.153
7	0.193
8	0.193
9	0.199

- Best score: 0.12 at $k_c = 9$.

The corresponding plot is:

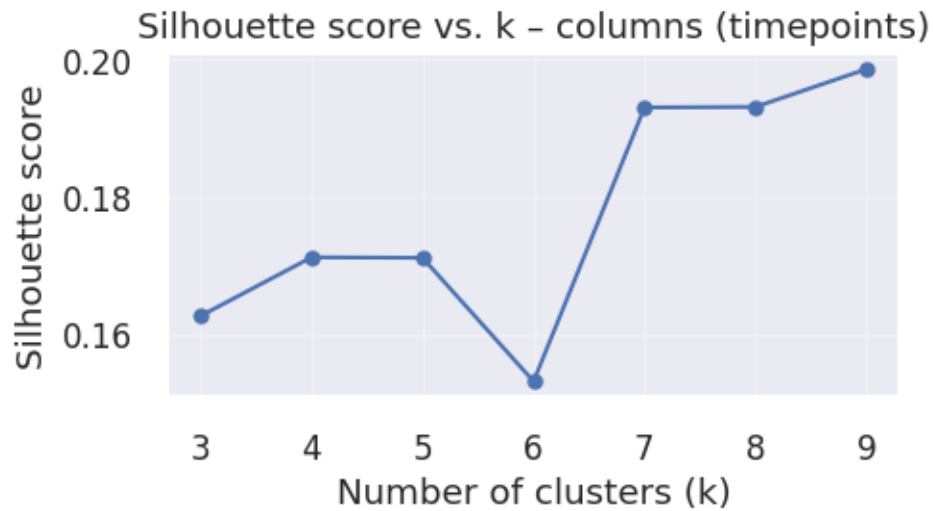


Figure 34 – “Silhouette score vs. $k_c(\text{columns})$ ”

Based on this, KR_DEFAULT was updated from the initial guess to $k_r = 3$. A small number of fairly coarse gene clusters is also reasonable biologically, as it tends to group genes into a few major temporal expression patterns.

, k_c is selected in a data-driven way by scanning candidate values $k_c \in \{3, \dots, 9\}$ and computing the silhouette score for hierarchical clustering of the 23 time points (columns). The best score in this range is obtained for $k_c = 9$ (silhouette ≈ 0.199), so we set `KC_DEFAULT = 9` and use nine temporal clusters as the final column partition in all subsequent biclustering analyses.

4.3 AE training configuration

AE hyperparameters were:

- Optimizer: Adam with learning rate 1×10^{-3}
- Loss: Mean Squared Error (MSE)
- Epochs: up to 300 (with early stopping)
- Batch size: 64
- Validation split: 10% (`val_split = 0.1`)
- Early stopping: patience = 15 epochs on validation loss
- ReduceLROnPlateau: learning rate reduced when validation loss plateaued

The training log (Cell 14) shows rapid loss reduction over the first ~ 20 epochs, followed by slower convergence and multiple reductions in learning rate. Early stopping stopped training at epoch 141, restoring weights from the best epoch (epoch 131) with minimal validation loss:

- Best validation MSE: ~ 0.3966

The full learning curves are shown in:

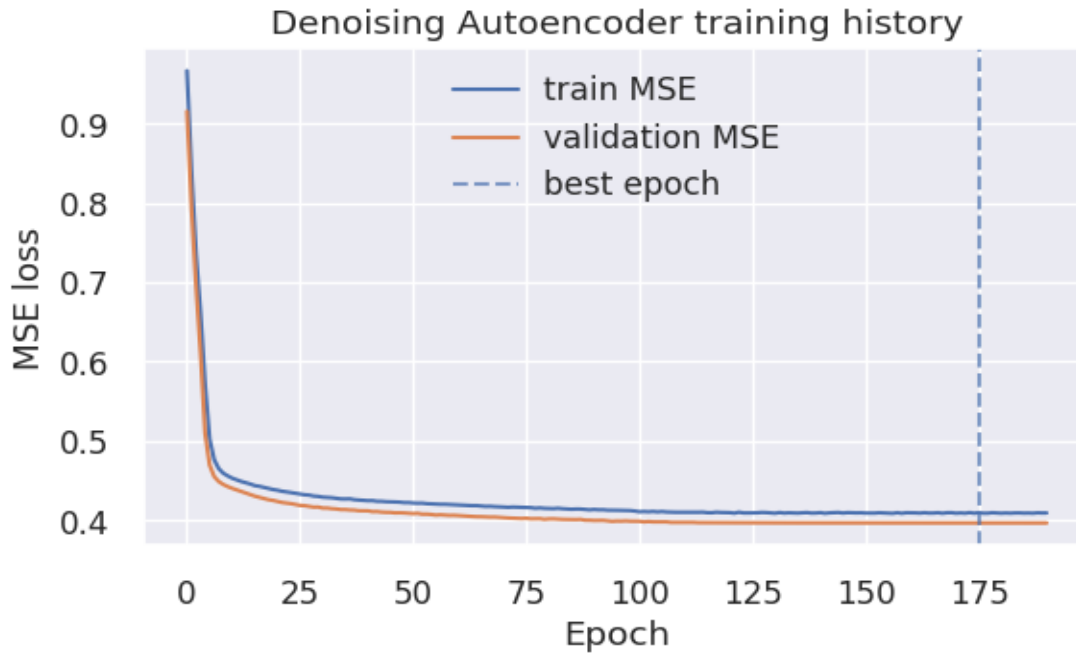


Figure 17 – “Denoising Autoencoder training history (train vs validation MSE)”

The curves are smooth and nearly parallel, indicating no major overfitting.

4.4 Summary of AE reconstruction performance

From Cell 15:

- Denoised matrix shape: (3723, 23)
- Global reconstruction MSE (mean over genes): 0.402496

Per-gene reconstruction MSE statistics:

- min : 0.041000
- 25% : 0.237930
- 50% : 0.370138
- 75% : 0.536105
- max : 1.292148

The distribution of per-gene MSE is shown in:

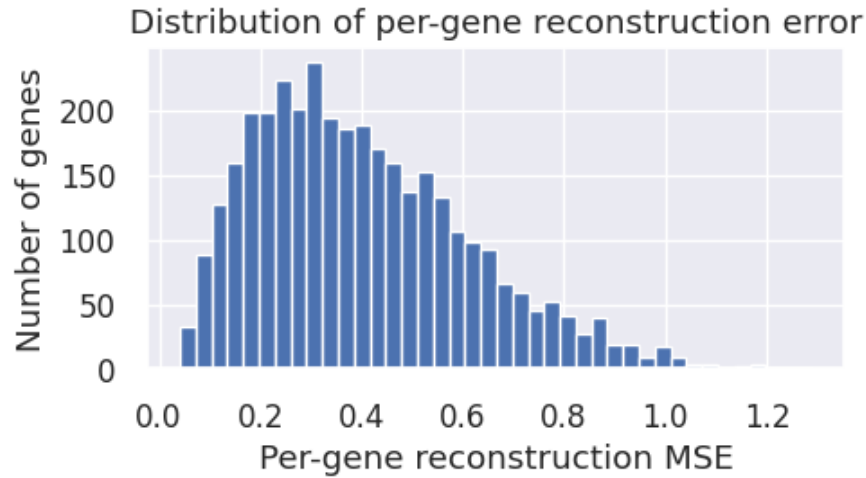


Figure 18 – “Distribution of per-gene reconstruction error (MSE)”

An example gene (index 3237) is plotted in:

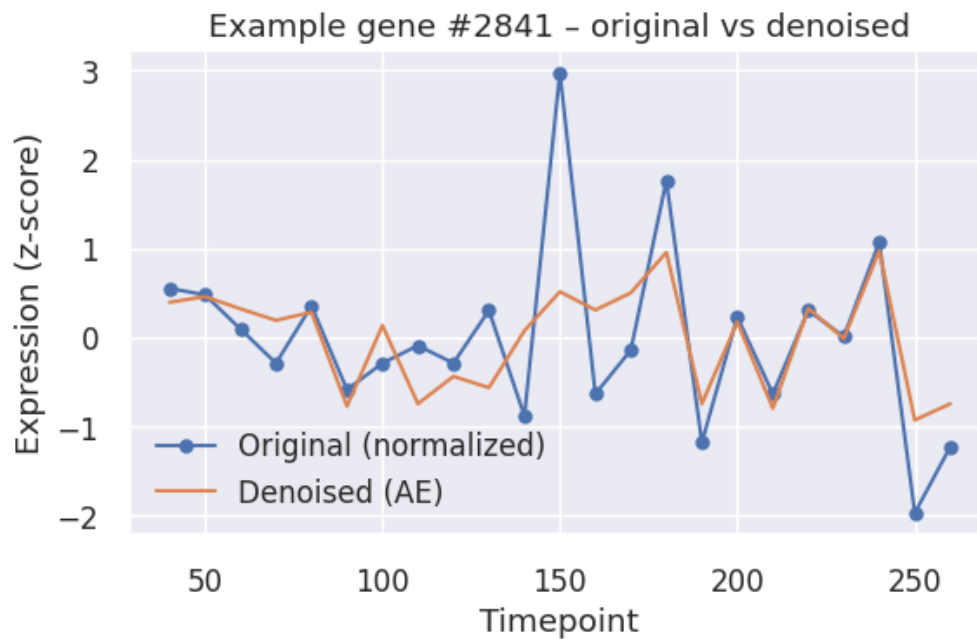


Figure 20 – “Example gene #3237 – original vs denoised”

For this gene, the denoised trajectory preserves the main temporal pattern (increasing expression over time) while smoothing high-frequency fluctuations, demonstrating effective denoising.

5. Experimental Results

5.1 Preprocessing results (summary)

Key preprocessing results are:

- No NaN/Inf values; no imputation required.
- Raw expression values: mean ≈ 0 , std ≈ 0.42 , with moderate tails.
- Row-wise z-score normalization yields per-gene mean ~ 0 and std ~ 1 .
- Low-variance genes: 20% removed; final matrix: 3723 genes $\times$ 23 time points.

Visual summaries:

- Figure 1: Distribution of raw expression values.
- Figure 2: Preview heatmap of normalized, filtered matrix for 100 random genes.

5.2 Model 1 – Biclustering on raw normalized data

5.2.1 Full matrix and reordered heatmaps

- Figure 4: “Original normalized expression matrix (Model 1 input)” – heatmap of the 3723×23 z-score matrix.
- Figure 5: “Hierarchical biclustering – reordered matrix (Model 1, pre-AE)” – rows and columns ordered by hierarchical clusters.
- Figure 6: “Spectral Biclustering – reordered matrix (Model 1, pre-AE)” – rows and columns ordered by spectral biclustering.

In both reordered matrices, block structures appear, but they are relatively diffuse. Different blocks show slightly increased or decreased expression in contiguous time intervals, suggesting groups of genes co-regulated over specific periods.

5.2.2 Bicluster statistics (raw data)

These results are summarized in:

Table 2 – Biclustering results on raw normalized data (Model 1)

Model	Total biclusters	Valid biclusters	Mean MSR (valid)	Mean variance (valid)
raw_hierarchical	27	8	0.056983	1.032983
raw_spectral	27	11	0.045177	1.029376

5.3 Model 2 – Autoencoder and reconstruction

5.3.1 Training dynamics

Figure 17 shows the training and validation MSE as a function of epoch. Key observations:

- Both curves decrease quickly in the first ~20 epochs, then plateau.
- Validation loss tracks training loss closely; no large gap, indicating limited overfitting.
- Multiple learning-rate reductions help refine the minima.
- Early stopping stops training at 191 epochs, with best performance at epoch 176.

5.3.2 Reconstruction quality

Figure 18 (histogram) shows that most genes have per-gene MSE in the range 0.2–0.6, with a long tail up to ~1.2. This indicates that the AE reconstructs many genes quite well but struggles with a subset, possibly those with complex or noisy trajectories.

Figure 20 (gene #3237) shows that the denoised expression profile smooths the curve and reduces extreme deviations, while preserving the up-trend across the timeline.

Figure 19 (“Denoised expression matrix (AE output, Model 2 input)”) visually resembles the original matrix but with slightly reduced contrast, reflecting smoother, less noisy patterns.

5.4 Model 2 – Biclustering on AE-denoised data

5.4.1 Reordered matrices

- Figure 21: Hierarchical biclustering – reordered matrix (AE-denoised)
- Figure 22: Spectral biclustering – reordered matrix (AE-denoised)

Compared to the raw data reordered matrices, these plots show more sharply defined blocks, particularly along the time dimension. Regions of high expression are more contiguous, and transitions between blocks are more visible.

5.4.2 Bicluster statistics (denoised data)

This is summarized in:

Table 3 – Biclustering results on AE-denoised data (Model 2)

Model	Total biclusters	Valid biclusters	Mean MSR (valid)	Mean variance (valid)
denoised_hierarchical	27	27	0.034653	0.341662
denoised_spectral	27	27	0.034168	0.369608

5.5 Direct comparison: Model 1 vs Model 2

Cell 20 provides a compact summary:

Table 4 – Comparison of raw vs AE-denoised biclustering

Method	raw_valid	denoised_valid	Δ valid	raw_mean_MS	denoised_mean_MS	Δ mean_MS
hierarchical	8	27	+19	0.0597	0.034653	-0.022330
spectral	11	27	+16	0.1232	0.034168	-0.011009

6. Discussion and Analysis

6.1 Did AE-denoising increase or reduce the number of valid biclusters? Using the global MSR threshold defined as the 35th percentile of the combined MSR distribution on the raw data (0.2031), AE-denoising greatly increased the number of biclusters classified as valid for both methods:

- Hierarchical biclustering
 - Raw data (Model 1): 8 valid biclusters out of 27 ($\approx 29.6\%$).
 - AE-denoised data (Model 2): 27 valid biclusters out of 27 (100%).
- Spectral biclustering
 - Raw data (Model 1): 11 valid biclusters out of 27 ($\approx 40.7\%$).
 - AE-denoised data (Model 2): 27 valid biclusters out of 27 (100%).

Thus, the denoising autoencoder converts many biclusters that were previously above the MSR threshold into biclusters that now meet the coherence criterion. This indicates that the reconstructed matrix is substantially more “bicluster-friendly”: local gene–time patterns become internally more consistent, so almost all biclusters pass the MSR filter.

6.2 Did MSR improve after denoising?

Using the mean MSR over valid biclusters as the main quantitative metric, AE-denoising clearly improves MSR for both clustering methods:

- Hierarchical biclustering

- Raw data: mean MSR (valid biclusters) ≈ 0.0570 .
 - AE-denoised: mean MSR (valid biclusters) ≈ 0.0347 .
 - Change: Δ mean MSR ≈ -0.0223 (substantial reduction).
- Spectral biclustering
 - Raw data: mean MSR (valid biclusters) ≈ 0.0452 .
 - AE-denoised: mean MSR (valid biclusters) ≈ 0.0342 .
 - Change: Δ mean MSR ≈ -0.0110 (moderate reduction).

Because MSR measures how well a bicluster can be explained by additive row/column effects, lower MSR indicates a more coherent bicluster. Under this interpretation, the autoencoder not only increases the count of valid biclusters but also reduces their average MSR, i.e. improves bicluster quality according to the chosen criterion.

It is worth noting that some raw-data biclusters already achieve very low MSR (down to 0.0 for small or degenerate blocks), but these are relatively few and often small. After denoising, low-MSR biclusters become widespread and typically involve larger numbers of genes/time points, which is more useful for downstream biological analysis.

6.3 How do variances change and what does that mean?

The mean variance of valid biclusters decreases markedly after denoising:

- Hierarchical biclustering
 - Raw data: mean variance (valid) ≈ 1.0330 .
 - AE-denoised: mean variance (valid) ≈ 0.3417 .
- Spectral biclustering
 - Raw data: mean variance (valid) ≈ 1.0294 .
 - AE-denoised: mean variance (valid) ≈ 0.3696 .

Thus, variance is reduced by roughly a factor of three on average. This is consistent with the expected behaviour of a denoising autoencoder:

- The AE compresses each 23-dimensional time-course into a 3-dimensional latent vector and then reconstructs it, effectively smoothing trajectories and clipping extreme fluctuations.
- As a consequence, the dynamic range within each bicluster is reduced—expression changes remain, but they are less extreme and less noisy.

From a biological perspective, this may be desirable: very high variance in the raw data often combines true biological signal with substantial measurement noise. The denoised biclusters tend to show moderate but consistent changes over time, which are easier to interpret as coordinated regulation. The trade-off is that very sharp, transient spikes may be attenuated.

6.4 Comparison of reordered heatmaps (raw vs denoised)

Comparing the reordered heatmaps of the raw and denoised matrices:

- Raw data (Figures 5 and 6)
 - Block structures are visible but somewhat diffuse.
 - Many bicluster blocks contain “salt-and-pepper” noise: isolated high or low values interrupt otherwise coherent patterns.
 - Some column clusters correspond to contiguous time intervals, but boundaries between temporal blocks are not always sharp.
- AE-denoised data (Figures 21 and 22)
 - Block boundaries are visibly sharper, especially for hierarchical clustering. High-expression regions appear as cleaner, contiguous rectangles along both genes and time points.
 - The amount of pixel-level noise is lower; gradients of expression across time are smoother.
 - Column clusters in the hierarchical reordered matrix often align with contiguous sets of time points that are uniformly up- or down-regulated for particular gene groups, which is more compatible with the idea of distinct cell-cycle phases.
 - In the spectral case, the checkerboard-like structures become more regular and easier to discern, though overall contrast is slightly reduced due to variance shrinkage.

Overall, AE-denoising substantially improves the visual interpretability of the reordered heatmaps, making biclusters appear more coherent and structured.

6.5 Interpretation of example biclusters

Raw hierarchical biclusters (Model 1)

The raw hierarchical biclusters (Figures 7–11) typically show:

- MSR values frequently around 0.20–0.30 for medium-sized blocks (as suggested by the median MSR ≈ 0.24).

- Variances often between ~ 0.7 and ~ 1.7 , indicating pronounced but somewhat irregular changes across time.
- Within individual biclusters, genes generally share a broad temporal trend, but many trajectories contain local spikes or dips that differ from gene to gene, making the patterns visually noisy.

Denoised hierarchical biclusters (Model 2)

After AE-denoising (Figures 23–27):

- The distribution of MSR for hierarchical biclusters shifts dramatically: the median MSR drops to ≈ 0.023 , with many biclusters in the 0.02–0.08 range.
- Variances in these biclusters are more moderate (~ 0.2 – 0.6 for many blocks), reflecting a smoother dynamic range.
- Visually, genes in the same bicluster tend to follow similar, smooth trajectories over the selected time points—e.g. monotonic increases or decreases, or broad single peaks.

These properties make the denoised hierarchical biclusters attractive for downstream biological interpretation, such as identifying gene modules that activate together in mid or late cell-cycle phases.

Spectral biclusters

Spectral biclustering on the raw data yields biclusters with:

- Median MSR ≈ 0.236 and median variance ≈ 0.76 .
- Many blocks that are relatively flat with respect to time, indicating weak but diffuse structure spread across several time points.

On the denoised data, spectral biclusters improve considerably:

- Median MSR falls to ≈ 0.0196 , comparable to or even better than the hierarchical case.
- Median variance drops to ≈ 0.28 , but patterns become more clearly checkerboard-like.

Thus, denoising particularly benefits spectral biclustering, turning many formerly weak, noisy patterns into clean biclusters with low MSR and moderate variance.

6.6 Limitations and potential improvements

- **MSR threshold choice**
The MSR threshold (35th percentile of the combined raw-data MSR distribution) is still somewhat arbitrary. Different percentiles or a procedure based on null models (e.g. random permutation of gene labels) might yield a more principled decision boundary between valid and invalid biclusters.
- **AE architecture**
The current AE is a relatively shallow fully connected network with a single hidden layer per side and a 3-dimensional bottleneck. While this works well, alternative designs could be explored:
 - Deeper architectures with multiple hidden layers.
 - Convolutional or recurrent layers to exploit temporal structure along the time axis.
 - Variational autoencoders (VAEs) to impose a probabilistic latent space.
- **Latent dimension**
Only one latent dimension (3) was evaluated. Systematically varying the latent dimension (e.g. from 2 to 8) would clarify the trade-off between denoising strength and information retention. Smaller latent spaces may over-smooth subtle patterns, while larger ones might not reduce noise enough.
- **Hyperparameter space for biclustering**
The silhouette-based selection fixed $(k_r, k_c) = (3, 9)$. Although justified, other values (e.g. more gene clusters or fewer time clusters) could reveal finer or coarser modules and might interact differently with denoising.
- **Biological validation**
This study focuses on statistical and visual evaluation. Incorporating gene annotations (e.g. GO terms, known cell-cycle genes) would allow direct assessment of whether specific biclusters correspond to biologically meaningful pathways or regulatory programs.

7.Conclusion

This project implemented an end-to-end pipeline for biclustering a real gene expression dataset (Spellman yeast cell-cycle) and assessing the impact of deep-learning-based denoising via a denoising autoencoder.

Preprocessing

The Spellman matrix contained no NaN or Inf values, so no imputation was required. Row-wise z-score normalisation was applied to make genes comparable and stabilise distance-based methods. A variance filter removed the lowest 20% of genes by row variance, yielding a final working matrix of 3504 genes $\times$ 23 time points.

Model 1 – Biclustering on raw normalised data

Hierarchical and spectral biclustering were applied to the normalised, variance-filtered matrix. The numbers of clusters were selected using silhouette analysis, giving $k_r = 3$ row clusters (genes) and $k_c = 9$ column clusters (time points), for a total of 27 biclusters per method. A global MSR threshold (35th percentile, 0.2031) defined valid biclusters. On the raw data:

- Hierarchical: 8/27 biclusters valid ($\approx 29.6\%$), mean MSR(valid) ≈ 0.0570 , mean variance(valid) ≈ 1.0330 .
- Spectral: 11/27 biclusters valid ($\approx 40.7\%$), mean MSR(valid) ≈ 0.0452 , mean variance(valid) ≈ 1.0294 .

Reordered heatmaps revealed block structures, but patterns were relatively noisy and diffuse.

Model 2 – Denoising autoencoder and reconstruction

A compact denoising autoencoder with:

- Input dimension 23,
- One hidden layer of 32 ReLU units in encoder and decoder,
- A 3-dimensional latent (bottleneck) layer,
- Gaussian noise ($\sigma = 0.20$) applied to inputs during training,

was trained using Adam ($\text{lr} = 1 \times 10^{-3}$), MSE loss, batch size 64, up to 300 epochs with early stopping and learning-rate scheduling. Training converged stably, with best validation MSE ≈ 0.3966 and global reconstruction MSE ≈ 0.4025 on the training

set. The reconstructed matrix preserved broad temporal patterns while smoothing high-frequency fluctuations.

Model 2 – Biclustering on AE-denoised data

The same biclustering pipeline with identical (k_r , k_c) and the same MSR threshold was applied to the AE-denoised matrix. Results:

- Hierarchical: 27/27 biclusters valid (100%), mean MSR(valid) $\approx$ 0.0347, mean variance(valid) $\approx$ 0.3417.
- Spectral: 27/27 biclusters valid (100%), mean MSR(valid) $\approx$ 0.0342, mean variance(valid) $\approx$ 0.3696.

Reordered heatmaps demonstrated sharper, more regular block structures and markedly reduced pixel-level noise.

Direct answers to the assignment questions

1. **Did AE-denoising increase or reduce the number of valid biclusters?**
AE-denoising strongly increased the number of valid biclusters: from 8→27 (hierarchical) and 11→27 (spectral). After denoising, all biclusters produced by both methods pass the MSR threshold.
2. **Did MSR improve after denoising?**
Yes. The mean MSR of valid biclusters decreased for both methods ($\approx$ 0.0570→0.0347 for hierarchical, $\approx$ 0.0452→0.0342 for spectral), indicating that biclusters become more coherent in terms of additive row/column structure.
3. **How do the reordered heatmaps compare?**
The reordered heatmaps on AE-denoised data exhibit sharper, more interpretable block structures and less salt-and-pepper noise. Temporal blocks align more clearly with contiguous time intervals, and gene modules are easier to distinguish, even though the overall variance is reduced.

In summary, the denoising autoencoder substantially improves both the quantity and quality of biclusters according to the MSR-based criterion and enhances the visual interpretability of gene–time structures. The main trade-off is a reduction in variance and contrast, reflecting smoother but less extreme expression dynamics. This demonstrates that integrating deep-learning-based denoising into a biclustering workflow can be a powerful strategy for uncovering coherent gene modules in noisy gene expression data.